DS223 Marketing Analytics
Homework 1
Bass Model

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March 2026

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1 Introduction

Advancements in Large Language Models (LLMs) have been one of the biggest topics of discussion within the field of Artificial Intelligence (AI) due to their widespread use across different domains [6]. In particular, from the first to the second half of 2025, almost every country saw an increase in the proportion of people using AI [1].

The aim of this paper is to provide a comparative approach, grounded in the Bass Model, to analyze the adoption rates of DeepSeek-R1 and OpenAI GPT-o1. DeepSeek was a major player during early 2025; it held a 43% market share in Russia and an 89% market share in China, while maintaining high usage across various other regions [1].

This study focuses on the monthly usage of DeepSeek-R1 and OpenAI-o1, concentrating on the early stages of 2025 when these models were still at the frontier of the industry. In particular, the Bass model provides a good fit for o1 diffusion and an acceptable fit for R1, suggesting that while the methodology can be applied to infer significant results, integrating these methods with other data providers could contribute to a broader understanding of the market.

The structure of the paper is as follows: Section 2 covers the products at hand, diving into the technical aspects of each model and how they disrupted the AI ecosystem. Section 3 briefly touches on the time-series data used for the analysis. Section 4 focuses on the core fundamentals of the Bass model and our chosen loss function. Section 5 provides both quantitative and qualitative analysis of the model fit and forecasts. Finally, Section 6 summarizes the results, highlighting current inferences and potential areas for growth in future studies.

2 Chosen Product

This section covers the primary innovation under study as well as the alternative innovation chosen for comparison. Sections 2.1 and 2.2 detail the primary and alternative innovations, respectively, while Section 2.3 justifies their problem fit and provides a comparative overview.

2.1 Chosen Innovation (DeepSeek-R1)

The chosen innovation for this assignment is **DeepSeek-R1**, which was featured in the *TIME* Best Inventions of 2025 (hyperlink) [8].

From late 2024 to early 2025, Large Language Models (LLMs) began showing diminishing returns relative to compute and data requirements [3]. While the premise that "scaling compute leads to better performance" remains generally true, performance often scales logarithmically. Furthermore, while scaling helps improve accuracy for tasks of moderate complexity, it has historically struggled to scale effectively for more complex reasoning problems [4, 6]. A core reason for these limitations was an over-reliance on human-annotated datasets, which introduces several failure modes:

- annotation bias,
- constrained reasoning paths, and
- high data collection costs.

The first and third points are clear; however, it is particularly noteworthy that during supervised fine-tuning (SFT), LLMs were found to only reach a reasoning level roughly equivalent to that of human annotators, which inherently capped performance [4].

DeepSeek-R1 was revolutionary upon release due to its novel application of Reinforcement Learning (RL) during the training process. Specifically, the developers aimed to allow the model to discover its own "reasoning traces" autonomously. These traces were later identified at a macro-level, demonstrating that language models can act as proxies for reasoning by benefiting from a direct mapping of language to correctness [5]. Despite subsequent limitations found in these traces, this approach highlights the inherent capability of LLMs to act as optimizers without requiring intermediate reward signals [9].

Furthermore, DeepSeek-R1 was the first open-weight reasoning model trained with RL to achieve frontier-level performance. Due to its open-weights nature, it is known that the team utilized a Mixture of Experts (MoE) architecture. This architecture allows for significantly higher efficiency during inference, which explains the substantial price gap per million tokens between *DeepSeek-R1* and *GPT-o1*.

2.2 Alternative Innovation (GPT-o1)

The alternative innovation considered is **GPT-o1**, chosen due to its nearly simultaneous release (December 2024) and its shared objective of incentivizing reasoning traces in LLMs. Specifically, the OpenAI team utilized Reinforcement Learning (RL) during the post-training phase to achieve two key improvements:

- **Process supervision**: Encouraging intermediate signals through problem decomposition, which guides the model toward a correct final solution.
- **Test-time compute scaling**: Allowing the model to allocate more computational resources during inference to "think" through complex problems.

Notably, this approach resulted in a "hidden" chain of thought generated internally by the model; though this was monitored by OpenAI for safety and accuracy at the time of publication, it was not made directly visible to the user.

2.3 Market Impact

Both architectures sought to advance LLM capabilities, primarily focusing on reasoning tasks where text serves as a powerful reward proxy. While both models

were designed for global release, their respective structural advantages and cost profiles allowed them to capture different segments of the AI market.

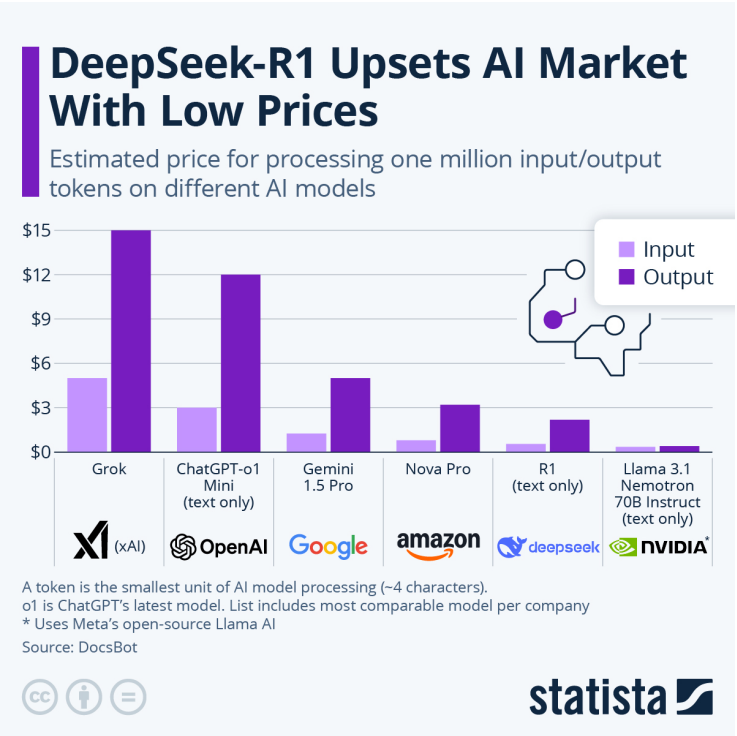

Figure 1: Comparison of input and output token pricing across AI models. Source: [2].

3 Historical Data

The **Similarweb "Gen AI Market Winners"** article utilizes **web traffic and engagement analytics** aggregated by Similarweb [7], (hyperlink). The dataset primarily consists of monthly website traffic estimates, supplemented by market share comparisons among leading generative AI platforms. Crucially, this data represents behavioral usage at the web and application levels rather than internal proprietary metrics, such as total tokens processed. Nevertheless, it serves as a reliable proxy for user demand, making it a suitable dataset for analyzing adoption trends and competitive positioning within the generative AI market.

Scope: Global. The traffic series is treated as a global adoption proxy, as the Similarweb report provides overall platform visit estimates without applying specific geographic filters for this analysis [7].

4 Estimating the Bass Model

The Bass diffusion model describes adoption driven by three primary factors:

1. **Innovation** (p): The likelihood of adoption due to external influence (advertising, news).
2. **Imitation** (q): The likelihood of adoption due to internal influence (word-of-mouth).
3. **Market Potential** (M): The total number of potential adopters in the market.

Let $A(t)$ be the cumulative number of adopters by time t, and $a(t)$ be the number of adopters in period t. Consequently, the cumulative and individual rates of adoption are defined as $F(t) = \frac{A(t)}{M}$ and $f(t) = \frac{a(t)}{M}$, respectively.

Solving the Bass model differential equation yields the cumulative distribution function:

$$F(t) = \frac{1 - e^{-(p+q)t}}{1 + \frac{q}{p} e^{-(p+q)t}}.$$

Given the discrete nature of the data, we model the number of adopters in period t, denoted as $\hat{a}_t$, as follows:

$$\hat{a}_t = \hat{A}(t) - \hat{A}(t-1) = M\left(F(t) - F(t-1)\right).$$

I estimate the parameters M, p, and q by minimizing the Sum of Squared Errors (SSE) between the observed monthly adopters and the Bass model predictions:

$$\arg\min_{M,p,q} \sum_{t=1}^{T} \left(a_t - \hat{a}_t(M,p,q)\right)^2.$$

The quality of the model fit is evaluated using the Root Mean Square Error (RMSE).

4.0.1 Fitted Models

DeepSeek: $\hat{M} = 7287.40, \quad \hat{p} = 0.06143, \quad \hat{q} = 0.07428.$
OpenAI: $\hat{M} = 111177.74, \quad \hat{p} = 0.03195, \quad \hat{q} = 0.14453.$
Here, $\hat{M}$ represents the market potential in millions of cumulative visits. For DeepSeek, $\hat{p}$ is relatively close to $\hat{q}$, suggesting that adoption was driven by both significant external interest and subsequent word-of-mouth. For the OpenAI (ChatGPT) proxy, $\hat{p}$ is notably lower than $\hat{q}$, indicating that growth is primarily driven by imitation and widespread social diffusion, consistent with prior observations [1].

While the classical Bass model assumes a smooth diffusion path, web traffic data often exhibits spikes due to exogenous events. This volatility can increase the error for series with sharp fluctuations. Nevertheless, the Bass model provides a robust first-order summary of diffusion and serves as a reliable basis for forecasting.

4.0.2 Look-alike Parameter Transfer

A parameter transfer experiment is also conducted. This involves adopting the innovation (p) and imitation (q) coefficients from the look-alike innovation (OpenAI) and re-estimating only the market potential (M) for DeepSeek to predict its adoption rate, $\hat{a}_t$.

5 Diffusion Model Analysis

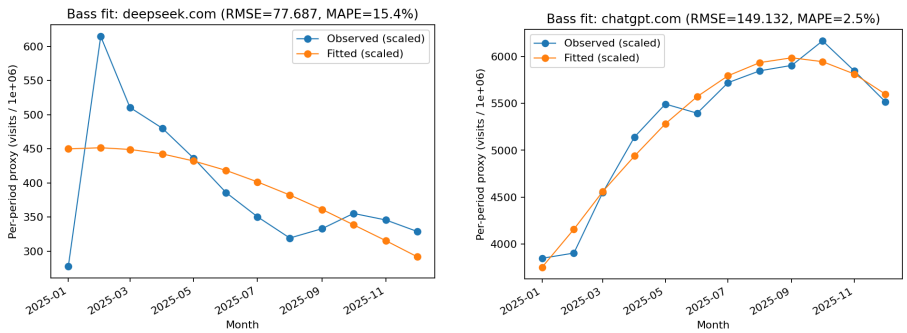

Figure 2: Monthly adoption proxy a_t (observed) and Bass fitted $\hat{a}_t$ (millions of visits).

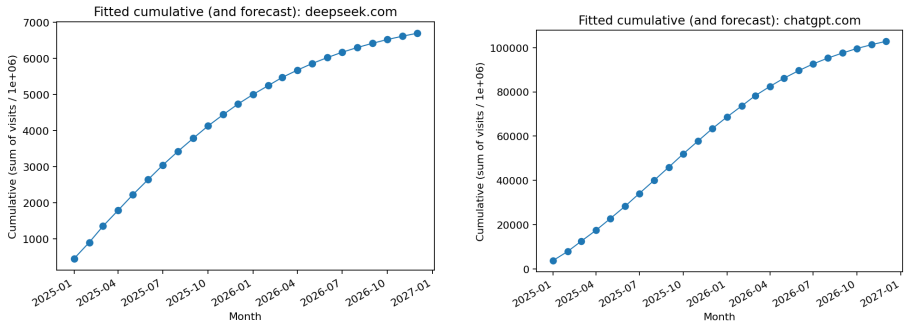

Figure 3: Cumulative Bass diffusion paths $\hat{A}(t)$ implied by the fitted models (millions of visits).

Exact monthly values and model outputs are reported in Appendix A, specifically in Tables 2 and 3.

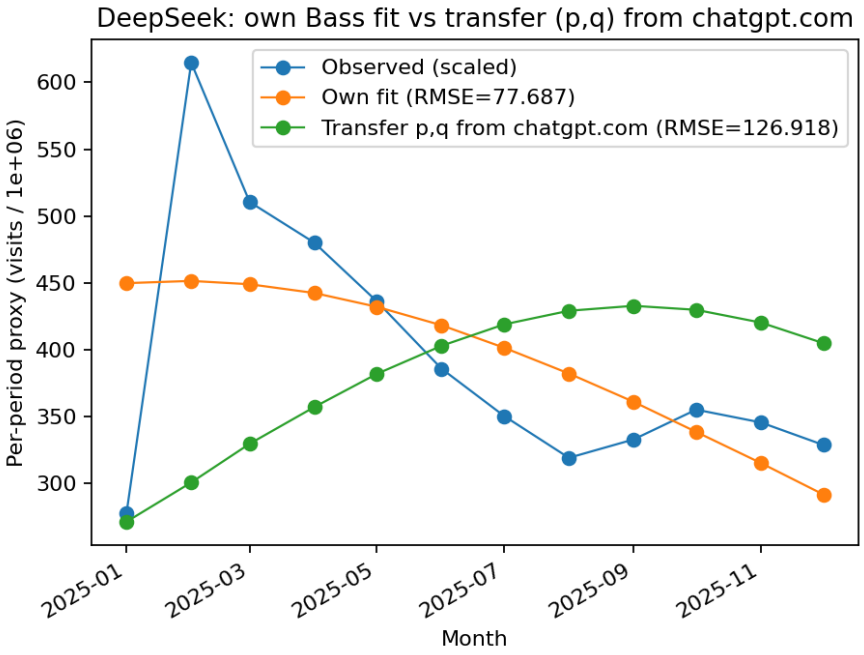

Figure 4: DeepSeek monthly adoption proxy a_t (observed) compared against: (i) DeepSeek's own fitted Bass model and (ii) a transfer model that reuses coefficients $(\hat{p}, \hat{q})$ from chatgpt.com while re-estimating M for DeepSeek. The transfer model exhibits higher error (RMSE), suggesting that DeepSeek's diffusion dynamics are distinct from those of GPT-o1.

6 Conclusion

6.1 Limitations

This study is subject to several limitations that should be noted. First, external factors such as sovereign AI usage tendencies and national safe-usage regulations were not considered due to the inherent ambiguity of these phenomena and the inability to provide exact measures or decay effects. Second, the broader impact on the AI hardware market, including key players such as NVIDIA, was not analyzed. Furthermore, while industry allegations have surfaced regarding DeepSeek's data collection methods in relation to OpenAI, these claims remain unproven and were therefore excluded to maintain the factual integrity of the study.

Another limitation involves the primary data source. While Similarweb provides a robust overview of the general market, access to a wider variety of data sources could lead to more precise estimates. Other data providers, such as OpenRouter and Statista (Source 1, Source 2), provide alternative perspectives

on daily and monthly usage; however, these were not utilized in the current analysis due to paywall constraints. Finally, the acquisition of high-frequency or "smoother" data would likely improve the convergence and overall fit of the Bass diffusion model.

A Appendix: Monthly adoption tables

A.1 OpenAI monthly table (own fit)

Table 1: OpenAI (chatgpt.com) monthly adoption proxy and Bass own-fit (millions of visits). Observed values are not available for forecast months.

month	Observed a_t	Fitted $\hat{a}_t$	Fitted $\hat{A}(t)$
2025-01	3849.850000	3753.640000	3753.640000
2025-02	3905.400000	4159.380000	7913.020000
2025-03	4548.610000	4559.430000	12472.450000
2025-04	5141.780000	4939.090000	17411.540000
2025-05	5492.390000	5282.190000	22693.730000
2025-06	5395.130000	5572.190000	28265.920000
2025-07	5720.230000	5793.750000	34059.670000
2025-08	5846.790000	5934.270000	39993.940000
2025-09	5904.120000	5985.360000	45979.300000
2025-10	6165.610000	5943.920000	51923.220000
2025-11	5844.550000	5812.480000	57735.700000
2025-12	5517.990000	5598.890000	63334.590000
2026-01	–	5315.380000	68649.970000
2026-02	–	4977.080000	73627.050000
2026-03	–	4600.470000	78227.520000
2026-04	–	4201.850000	82429.370000
2026-05	–	3796.080000	86225.440000
2026-06	–	3395.800000	89621.240000
2026-07	–	3011.010000	92632.250000
2026-08	–	2648.980000	95281.230000
2026-09	–	2314.450000	97595.680000
2026-10	–	2010.010000	99605.690000
2026-11	–	1736.500000	101342.190000
2026-12	–	1493.420000	102835.610000

A.2 DeepSeek monthly table (own fit)

Table 2: DeepSeek (deepseek.com) monthly adoption proxy and Bass own-fit (millions of visits). Observed values are not available for forecast months.

month	Observed a_t	Fitted $\hat{a}_t$	Fitted $\hat{A}(t)$
2025-01	277.910000	449.870000	449.870000
2025-02	614.740000	451.530000	901.400000
2025-03	510.560000	449.040000	1350.440000
2025-04	480.120000	442.510000	1792.950000
2025-05	436.200000	432.160000	2225.110000
2025-06	385.850000	418.360000	2643.470000
2025-07	350.480000	401.560000	3045.030000
2025-08	319.270000	382.300000	3427.320000
2025-09	333.000000	361.130000	3788.450000
2025-10	355.280000	338.640000	4127.100000

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Table 2: DeepSeek (deepseek.com) monthly adoption proxy and Bass own-fit (millions of visits). Observed values are not available for forecast months.

month	Observed a_t	Fitted $\hat{a}_t$	Fitted $\hat{A}(t)$
2025-11	345.710000	315.370000	4442.460000
2025-12	328.870000	291.810000	4734.270000
2026-01	–	268.410000	5002.680000
2026-02	–	245.540000	5248.220000
2026-03	–	223.500000	5471.720000
2026-04	–	202.510000	5674.220000
2026-05	–	182.730000	5856.950000
2026-06	–	164.270000	6021.220000
2026-07	–	147.180000	6168.400000
2026-08	–	131.480000	6299.880000
2026-09	–	117.130000	6417.010000
2026-10	–	104.100000	6521.110000
2026-11	–	92.330000	6613.440000
2026-12	–	81.730000	6695.160000

A.3 DeepSeek monthly table (transfer from chatgpt.com)

Table 3: DeepSeek monthly adoption proxy and Bass transfer fit (millions of visits). Transfer model reuses $(\hat{p}, \hat{q})$ from chatgpt.com and re-estimates M for DeepSeek. Observed values are not available for forecast months.

month	Observed a_t	Transfer $\hat{a}_t$	Transfer $\hat{A}(t)$
2025-01	277.910000	271.490000	271.490000
2025-02	614.740000	300.840000	572.330000
2025-03	510.560000	329.770000	902.100000
2025-04	480.120000	357.230000	1259.330000
2025-05	436.200000	382.050000	1641.380000
2025-06	385.850000	403.020000	2044.400000
2025-07	350.480000	419.050000	2463.440000
2025-08	319.270000	429.210000	2892.650000
2025-09	333.000000	432.900000	3325.560000
2025-10	355.280000	429.910000	3755.460000
2025-11	345.710000	420.400000	4175.860000
2025-12	328.870000	404.950000	4580.820000
2026-01	–	384.450000	4965.260000
2026-02	–	359.980000	5325.240000
2026-03	–	332.740000	5657.980000
2026-04	–	303.910000	5961.890000
2026-05	–	274.560000	6236.450000
2026-06	–	245.610000	6482.060000
2026-07	–	217.780000	6699.840000
2026-08	–	191.590000	6891.430000
2026-09	–	167.400000	7058.830000
2026-10	–	145.380000	7204.210000
2026-11	–	125.600000	7329.800000
2026-12	–	108.020000	7437.820000

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